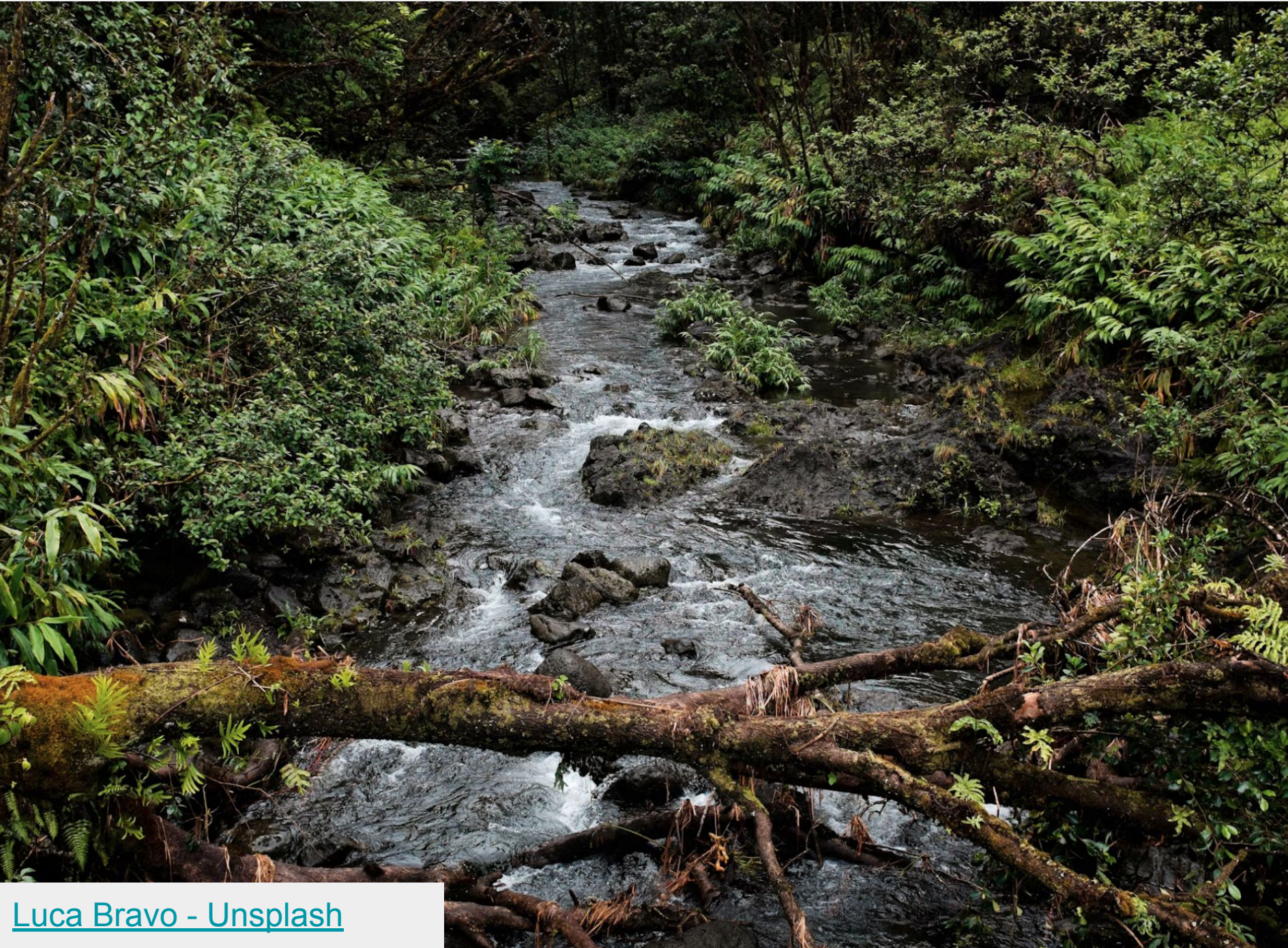


sed

Adapted from materials by Dr. Carrier



sed

Stream EDitor

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Stream can be a file or a piped input

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Sstream EDitor

Stream can be a file or a piped input

General form:

```
sed [options] '/pattern/ command' input_file
```

```
sed [options] 'range command' input_file
```

Common options

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-E to use extended regex

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-f filename to load sed commands from a file

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`-E` to use extended regex

`-f` filename to load sed commands from a file

`--in-place` or `-i` to modify a file in place

`--in-place=suffix`

Or

`-isuffix`

To create a backup file with suffix extension

Stream manipulation

Deleting lines:

- `sed '4 d'`

- `sed '1,4 d'`

- `sed -E '/pattern/ d'`

(\$ = end of file when using ranges)

Stream manipulation

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- `sed '4 d'`
- `sed '1,4 d'`
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Substituting text:

- `sed -E 's/pattern/replace_str/'`

Stream manipulation

Deleting lines:

- `sed '4 d'`
- `sed '1,4 d'`
- `sed -E '/pattern/ d'`

(\$ = end of file when using ranges)

Substituting text:

- `sed -E 's/pattern/replace_str/'`
- Example:
`sed -E 's/ *$// ' filename`

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
```

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

 - `sed -E 's/ *$//' filename`

- By default, only replaces first match per line

 - Replace all:

 - `sed -E 's/pattern/replace_str/g'`

 - Replace second:

 - `sed -E 's/pattern/replace_str/2'`

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
```


Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
```

- Can combine patterns with ranges:

- `sed -E '5,$ s/pattern/replace_str/'`

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
```

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
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- We can also use parts of the matched pattern in our replacement! Uses (parentheses groupings)

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
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- We can also use parts of the matched pattern in our replacement! Uses (parentheses groupings)
- To refer to parentheses groups: \1, \2, ...

- `sed -E 's/an? ([[:alpha:]]+) /\1/'`

Text substitution

Substituting text:

- `sed -E 's/pattern/replace_str/'`

- Example:

```
sed -E 's/ *$//' filename
```

- We can also use parts of the matched pattern in our replacement! Uses (parentheses groupings)
- To refer to parentheses groups: \1, \2, ...

- `sed -E 's/an? ([[:alpha:]]+) /\1/'`

- To refer to the whole matched pattern: &

- `sed -E 's/(\.+\.com)/www.&/'`

Challenge question

How can we take a list of movies, and for any movie that is of the form:

"The XYZ"

we change it to

"XYZ, The"